

Zero-Knowledge Proof Framework for Identity Verification and Interoperable Payments on the Decentralized Web

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Digital identity verification is central to trust management on the evolving decentralized web. Traditional web-based identity models, which are heavily centralized and dependent on trusted intermediaries, pose significant challenges related to user privacy, data security, and regulatory compliance, especially in sensitive contexts such as Know Your Customer (KYC) processes. This paper introduces a novel privacy-preserving KYC verification framework leveraging Zero-Knowledge Proofs (ZKPs), Self-Sovereign Identity (SSI), Decentralized Identifiers (DIDs), and smart contracts, explicitly designed as a decentralized trust infrastructure for web-based interoperable payments. Our approach enables users to verify their identities across multiple platforms without revealing sensitive personal information, thereby significantly reducing long-term reliance on centralized authorities and enhancing user control and privacy. Furthermore, our system achieves cross-chain interoperability, ensuring that identity verification credentials can be securely and efficiently recognized across diverse Web3 ecosystems. We present a detailed prototype implementation of our DID framework, highlighting its ability to meet regulatory requirements while ensuring seamless interoperability across platforms. Comprehensive performance evaluations, including metrics on proof generation time, gas consumption, and transaction costs, demonstrate that the framework achieves low-latency verification and efficient execution, making it suitable for high-throughput, web-scale deployment.

CCS Concepts: • **Security and privacy** → **Privacy-preserving protocols**; **Digital signatures**; **Distributed systems security**; • **Information systems** → *Web applications*; **Electronic funds transfer**; *Secure online transactions*; • **Networks** → *Security protocols*; • **Applied computing** → **Digital cash**; • **Software and its engineering** → **Distributed systems organizing principles**;

Additional Key Words and Phrases: Zero-Knowledge Proofs, Self-Sovereign Identity (SSI), Decentralized Identifiers (DIDs), Verifiable Credentials, Web3 Identity, Smart Contracts, Web3, Regulatory Compliance (KYC/AML), Privacy-Preserving Compliance, Cross-Border Payments, Blockchain Interoperability, Blockchain-Based Trust Infrastructure, Regulatory Technology

1 Introduction

In today's rapidly evolving digital environment, traditional web-based identity verification methods face growing challenges in balancing user privacy, security, and regulatory compliance. Because they are predominantly centralized, these identity management solutions rely on trusted third parties such as financial institutions, service providers, and platform operators. This model requires individuals to repeatedly share sensitive personal information, which is then stored on centralized silos that significantly increase the risk of data breaches, identity theft, and unauthorized data exposure. Furthermore, reliance on multiple centralized intermediaries creates inefficiencies, inflating transaction costs, increasing processing delays, and complicating secure data management. Such limitations are emblematic of a broader tension within the Web's evolution.

Previous web-based identity systems have consolidated user data into siloed, proprietary platforms, limiting portability and user control. In contrast, the decentralized web, often referred to as Web3, is reimagining digital

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identity infrastructure based on principles of Self-Sovereign Identity (SSI), interoperability, and cryptographic trust. Blockchain wallets, Decentralized Identifiers (DIDs), and Verifiable Credentials (VCs) enable users to assert facts about themselves without surrendering raw data to identity verifiers, which is a core tenet of the Web3 vision. However, while Web3 has made significant strides in establishing DID primitives, it now faces its next major challenge: reconciling privacy with regulatory compliance in sensitive domains such as finance, payments, and government services.

The tension between decentralization and trust reflects a broader challenge for the Web: how to support accountability without reintroducing centralized control especially for personal data [61]. Prior work has argued that accountability should be treated as a first-class property of Web systems, enabling actions and decisions to be traced, verified, and audited without requiring unconditional trust in intermediaries [69, 73]. Our work builds on this vision by operationalizing accountability for identity verification and cross-border payments, enabling users to cryptographically demonstrate claims such as “I am verified” and “I have sufficient funds” without exposing any private data. By leveraging SSI, DIDs, and ZKPs, our framework shifts verification from centralized gatekeepers and other single points of control to decentralized cryptographic protocols. In other words, unlike traditional web identity systems, where banks or platforms hold and mediate access to user data, our framework places users in full control of their digital credentials. Users share proofs, not personal information, supporting the decentralized web’s goal of empowering individuals and minimizing surveillance.

Ultimately, our framework contributes to a more user-centric web where individuals retain control over their digital identities and share verifiable proofs instead of raw data, thereby bridging the gap between regulatory assurance and user privacy.

Contributions

We present a decentralized, privacy-preserving identity and compliance verification framework for the decentralized web, designed to support scalable, trust-minimized interaction across different blockchain-based systems and traditional web applications. Our approach combines ZKPs, SSI, and smart contract-based credential verification to let users prove regulatory compliance requirements such as, Know Your Customer (KYC), Anti-Money Laundering (AML), and Counter-Terrorism Financing (CTF), without disclosing sensitive personal information. This design advances decentralized trust infrastructure by reconciling privacy with auditability, providing a deployable foundation for secure identity verification and regulatory assurance in cross-border, cross-platform settings.

The key contributions of this work are:

(1) **A DID Framework for Privacy-Preserving Compliance.**

We propose **zkKYC**, a DID verification mechanism that allows users to demonstrate compliance with KYC without exposing underlying personal data. By combining DIDs, VCs, and ZKPs, our framework enables user-controlled, privacy-preserving authentication suitable for decentralized web applications. This advances the vision of a user-centric web by replacing data disclosures with verifiable proofs.

(2) **A Cross-Platform Protocol for Interoperable, Trust-Minimized Payment and Reporting.**

We introduce **zkCBPR**, a cryptographic payment and regulatory Cross-Border Payments and Reporting (CBPR) protocol that extends zkKYC to support transactions across heterogeneous blockchain networks and web-based payment channels. The zkCBPR framework promotes cross-chain interoperability by enabling verification of credentials and compliance proofs across different ledgers without exposing sensitive transaction history. This supports decentralized finance (DeFi) and other Web3 ecosystems where regulatory compliance must coexist with privacy and platform diversity.

(3) **End-to-End System-Level Evaluation of the Full Pipeline.**

We provide an end-to-end, system-level evaluation of the complete zkKYC/zkCBPR pipeline, executing

the workflow from DID issuance and on-chain registration, through transaction certificate issuance and validation, to ZKP generation/verification for KYC compliance and fund sufficiency, and finally on-chain payment execution and Merkle Tree Root (MTR) update. Even under an emulated deployment in which protocol roles are instantiated as separate components, we report a phase-by-phase latency breakdown. This analysis highlights the dominant cost of local ZKP generation, in contrast to the low latency of verification and settlement. Together, these results demonstrate the feasibility of zkKYC/zkCBPR as an integrated protocol rather than a collection of isolated cryptographic primitives.

2 Background and Related Work

The decentralized web introduces new paradigms for digital identity, trust, and compliance, moving away from centralized control and toward user-centric, cryptographically verifiable systems. Our work intersects with several active research and implementation areas, including DID infrastructures, privacy-preserving credential verification mechanisms, and cross-chain interoperability. In this section, we contextualize our contributions within these domains, highlighting foundational standards and comparing them to relevant academic and industry efforts.

2.1 Decentralized Web and Blockchain

The decentralized web aims to create a more open, transparent, and user-centric ecosystem by leveraging blockchain technology, which acts as the foundational trust layer [36, 72]. A blockchain is a distributed ledger designed to be append-only, and records transactions and data across many computers without a central intermediary [60]. A blockchain's integrity and security are maintained through consensus mechanisms, such as Nakamoto consensus or Proof-of-Work (PoW) [60] and Proof-of-Stake (PoS) [49], where many independent participants agree on the current state of the ledger. This decentralized agreement makes the data tamper-resistant and publicly verifiable. Blockchain networks can be either permissionless (e.g., Bitcoin [60], Ethereum [21]), where anyone can join and participate, or permissioned (e.g., Hyperledger Fabric [4]), where participation is limited to known and vetted entities. Choosing between these models involves trade-offs among public auditability, privacy, scalability, and governance, and these choices directly shape the design of DID systems built on top of them. The introduction of smart contracts, pioneered by Ethereum [21], greatly expanded what blockchains can do beyond simple value transfer. A smart contract is a program stored on the blockchain that executes automatically in response to specified inputs or conditions. With smart contracts, developers can build decentralized applications (dApps) for many uses, including finance, governance, and most relevant to our work, *identity management* and *payments*. While blockchain offers a strong foundation for trustless systems, persistent challenges with scalability and interoperability across heterogeneous ledgers remain. Addressing these challenges has become a central focus of ongoing research and system design, efforts to which our work contributes by proposing a framework that reconciles privacy, compliance, and interoperability.

2.2 Know Your Customer (KYC) and Know Your Business (KYB)

KYC and KYB are regulatory compliance processes requiring financial institutions, cryptocurrency exchanges, and other service providers to verify the identity of individual clients (with KYC) and corporate entities (with KYB). Its main goal is to prevent illegal activities such as money laundering, terrorist financing, and fraud by ensuring that participants in financial transactions are properly identified. A typical KYC process includes three main steps: (1) collecting personally identifiable information (PII) such as name, date of birth, and address; (2) verifying these details against official documents like passports or ID cards; and, (3) in some cases, performing additional checks such as biometric verification, facial recognition, or cross-referencing with global watchlists. KYC requirements are mandated by regulations worldwide, such as the USA PATRIOT Act in the United States [84]

and the AML Directives in the European Union [28]. While these measures strengthen trust and compliance, they also raise concerns about user privacy, data security, and onboarding costs. Parallel requirements exist for corporate entities under KYB, which extends similar verification principles to businesses by confirming ownership structures, ultimate beneficial owners, and legal standing. Increasingly, both KYC and KYB processes involve issuing verifiable digital credentials that attest to successful identity verification, thereby improving efficiency in onboarding, strengthening risk management, and ensuring regulatory compliance while raising important questions around user privacy, data security, and operational costs.

2.3 Decentralized Identity (DID) and Trust Infrastructure

A key trend in the decentralized web is the shift from centrally-managed identities to SSI and DIDs. The W3C DID standard defines a format for persistent, verifiable digital identifiers that do not rely on any central authority [17]. Complementing DIDs, the W3C VCs data model [18] provides a way to express and cryptographically verify claims or credentials on the Web in an interoperable, privacy-respecting manner. Together, DIDs and VCs enable users to control their identity data and present tamper-evident credentials.

Real-world implementations of this paradigm have rapidly advanced. For example, the Sovrin network built on Hyperledger Indy provides a public-permissioned ledger for SSI, supporting decentralized trust through DIDs and cryptographic credentials with selective disclosure [59, 93]. The European Blockchain Services Infrastructure (EBSI), launched by the EU in 2018, is deploying a cross-border blockchain network to support government services with DIDs and VCs as core components, demonstrating the public sector’s commitment to DID [15]. In the private sector, Microsoft’s Project ION leverages the Bitcoin blockchain to implement a permissionless DID network using the Sidetree protocol, thereby anchoring identity proofs on a global public ledger without centralized validators [53]. These efforts share the goal of distributing the “root of trust” for identity: instead of a single provider (like a social login or certificate authority), trust is established via consensus across decentralized networks and cryptographic verification of credentials. However, challenges remain in bootstrapping trust and preventing abuses. For instance, one must ensure that each identity represents a real, unique user (avoiding Sybil attacks) and that lost private keys (which control DIDs) can be managed. Research projects like *CanDID* (Can-Do Decentralized ID) have explored incorporating legacy identity information and committee governance to achieve Sybil-resistant, accountable identity management without recentralizing trust [52]. Polygon ID (rebranded as Privado ID) [3] is a zero-knowledge native DID framework originally developed within the Polygon ecosystem. It allows users to generate and manage SSIs and VCs using ZKPs to prove identity attributes (e.g., age, KYC status) without revealing sensitive information. Designed for compatibility with W3C DIDs and VCs, Privado ID provides smart contract tooling to enable private, on-chain verification for Web3 applications such as token-gated access, DAO voting, and DeFi compliance workflows. Its extensibility across EVM and non-EVM chains positions it as a flexible identity solution for the decentralized web. In contrast, the KILT Protocol [68] is a Substrate-based blockchain project launched as a Polkadot parachain that focuses on issuing reusable VCs with full W3C DID and VC compliance. KILT emphasizes decentralized trust by allowing any party to act as an identity attester, and it supports privacy-preserving selective disclosure without relying on ZKPs. Notably, KILT has seen real-world deployment in collaboration with enterprise partners such as Deloitte [67], enabling regulatory-grade KYC/KYB credential issuance and on-chain identity assurance for traditional and decentralized markets. Both platforms illustrate key trends in DID: user control, verifiability, and interoperability, with differing approaches to trust models and privacy guarantees.

Our work builds on the foundations laid by these DID frameworks: we use DIDs as user identifiers in our system, enabling a user-controlled identity that can be recognized across platforms, and we leverage a blockchain-based registry (Merkle-tree anchored) to make these DIDs verifiable globally. In doing so, we contribute to a

decentralized trust infrastructure for the Web, wherein no single authority monopolizes identity verification and trust is rooted in cryptographic proofs and distributed consensus.

2.4 Zero-Knowledge Proofs (ZKPs) and Privacy-Preserving Credential Verification

As DID gains adoption, an equally important thread is privacy-preserving verification of credentials at web-scale. Traditional KYC/KYB processes require sharing extensive personal data with each service provider, which conflicts with the privacy ethos of SSI. Cryptographers have long envisioned proving facts about one’s identity without revealing the identity itself. Notably, ZKPs of identity were proposed as early as the 1980s [31]. In recent years, modern ZKP systems, such as Zero-Knowledge Succinct Non-Interactive Argument of Knowledge (zkSNARKs) [11], have become efficient enough for practical use on web- and blockchain-based platforms. Lavin *et al.* [50] survey a wide range of ZKP applications in blockchain and conclude that ZKPs are versatile tools for securing transactions and identity information. There is also research into ZKP-based credential schemes that can meet real-world regulatory requirements. For example, Constantinides and Cartledge [19] introduce BlockVerify, a framework on Ethereum where users can prove compliance with KYC/AML rules by presenting a ZKP of a credential (such as “I am over 18” or “I passed KYC with bank X”), all without exposing any personal details. BlockVerify demonstrated the viability of on-chain credential verification but was limited to a single blockchain ecosystem. Our approach generalizes this idea to work across multiple ledgers and contexts.

In parallel, the concept of “zkKYC” has been gaining traction. Pauwels *et al.* [64] first outlined zkKYC as an SSI solution for KYC in which users prove their identity attributes via ZKPs instead of disclosing raw identity credentials or official documents. They later implemented a prototype for DeFi platforms, showing that smart contracts and zkSNARKs can enforce KYC compliance in DeFi while preserving user anonymity [65]. We build directly on this line of work: our zkKYC/zkCBPR framework extends the zkKYC concept to the CBPR domain, i.e., our exemplar application, integrating it with payment workflows and adding support for interoperable credentials. In doing so, we address the open challenge of performing regulated identity checks in a decentralized manner on the Web. Notably, our design uses a combination of ZKPs, Merkle trees, and digital certificates to keep KYC data hidden from on-chain transactions yet verifiable by authorized parties. This approach aligns with the *minimal disclosure* principle advocated in SSI [2], and it echoes the *privacy-by-design* schemes proposed for digital currencies [27]. For instance, Pocher and Veneris [66] describe an AML/KYC compliance scheme for central bank digital currencies (CBDCs) that uses cryptographic commitments and ZKPs to balance transparency with privacy. Such schemes underscore that it is possible to satisfy regulators (e.g., proving a user is not a sanctioned individual) without revealing the user’s identity to every verifier. Furthermore, emerging “personhood” verification methods demonstrate the importance of privacy-preserving trust at web-scale: Adler *et al.* [1] introduce the notion of *personhood credentials*, which leverage AI and cryptographic tools to distinguish real humans from bots in online interactions while respecting privacy. This idea is complementary to zkKYC, as both aim to prove something about a user (be it proof of humanity or KYC status) with zero-knowledge guarantees. Our work unites these threads by enabling users to prove compliance (a proxy for trustworthiness in financial transactions) with strong privacy guarantees. We use state-of-the-art ZKP technology, which was made developer-friendly by languages like Circom [8], and design our zkKYC/zkCBPR framework to fully adhere to SSI standards.

Empirical studies of DeFi further highlight the dynamic nature of risk and compliance in Web3 systems. Recent work on DeFi survival analysis [33, 38–40] examines time-to-event behaviors such as liquidation and protocol exit, demonstrating that user risk profiles evolve continuously in response to market conditions. These findings underscore the limitations of static, one-time verification models and motivate compliance infrastructures that can support ongoing, verifiable assessment of user status without repeated disclosure of sensitive data. Our framework addresses this need by enabling recurrent, privacy-preserving verification of identity and transaction readiness through cryptographic proofs.

2.5 Cross-Chain Interoperability

An important, yet challenging, aspect of DID implementation is interoperability across different blockchain networks and existing web-based systems. Early research on blockchain applications in CBPR showed efficiency gains by removing intermediaries [22], and recent work has aimed to integrate blockchain solutions with the prevailing financial infrastructure. Mridul *et al.* [55] propose a smart contract-based CBPR framework that aligns transactions with the ISO 20022 banking standard for messaging [79], illustrating how decentralized systems can interface with traditional payment networks. These efforts, however, generally assume a unified environment or single-ledger context for identity and transactions. In practice, the decentralized web is heterogeneous with multiple blockchain platforms, such as Ethereum, Bitcoin, and consortium chains, as well as numerous identity providers or schemes that may need to interoperate [47]. Thus, our contribution in this regard is a *cross-chain capable* identity verification approach.

By using DIDs (which are URI-based and can incorporate methods for different ledgers) and by publishing credential proofs on a public blockchain, our framework makes it possible for a user's verified identity to be recognized across disparate dApps. For example, a user's DID and associated proof of KYC could be checked by a smart contract on an Ethereum sidechain or by a service on a Hyperledger Fabric network, so long as they trust the root blockchain where the DID and proof are anchored. This approach is inspired by recent interoperability solutions: Harmonia, for instance, uses ZKPs to enable secure communication between different blockchain networks without a centralized bridge [6]. While Harmonia focuses on general cross-chain security, we target cross-chain trust in the context of identity, ensuring that a credential issued on one platform can be verified on another. Likewise, Mohanty *et al.* [54] emphasize in their interoperability survey, that standard protocols and consensus on identity data are crucial for sustainable cross-chain payments. Our design addresses this by leveraging standard DID methods, therefore any platform that understands W3C DIDs can interpret the zkKYC/zkCBPR DIDs introduced in this paper and not tie the verification to any single proprietary system.

Finally, we note that “decentralizing trust” does not eliminate the need for governance; instead, it changes its nature. SSI ecosystems often introduce trust frameworks or governance authorities (like the Sovrin Foundation [77] or Trust Over IP Council [83]) to define rules for issuers and verifiers of credentials [93]. In our framework, the Certificate Authority (CA), which is typically operated by a trusted body such as a financial regulator, industry consortium, or government agency, plays a role in bootstrapping trust by validating real-world identity and issuing a DID. However, thereafter, trust is maintained in a decentralized fashion: the verifiers validate the attached proofs, and those proofs are valid across organizational boundaries. This aligns with real-world DID implementations such as *Hyperledger Indy/Sovrin* [59, 77] and *EBSI* [15], where a few authoritative parties issue credentials or DIDs under agreed policies, and any verifier can cryptographically confirm a credential's validity and integrity.

2.6 Related Application Domains

Beyond identity-specific solutions, other blockchain-based frameworks have explored how distributed ledger technologies can support data integrity, standardization, and verifiable trust in broader decentralized web applications.

Blockchain-based trust infrastructures have also been explored in other regulated, data-sensitive domains such as healthcare. For example, Li *et al.* [51] propose a standards-based ontological model for healthcare smart contracts, emphasizing semantic interoperability and data governance across distributed ledgers. Additionally, BlockIoT demonstrates how blockchain can be used to securely integrate and manage IoT-generated health data while preserving data integrity and access control [63, 74–76]. While their focus is on healthcare, their method of aligning smart contract logic with structured data models echoes our approach of integrating regulatory reporting structures with verifiable credential proofs.

Related work has also explored how domain-specific decision logic can be formalized and enforced on decentralized infrastructures. For example, recent work on translating clinical decision logic within knowledge graphs to smart contracts demonstrates how regulated, expert-defined rules can be compiled into executable, auditable blockchain programs [48, 85, 94]. While situated in the healthcare domain, this line of research highlights the broader feasibility of encoding compliance and decision logic as smart contracts, enabling consistent enforcement and verifiable execution across organizational boundaries. Our work similarly operationalizes regulatory logic, but focuses on identity verification and cross-border payments, integrating cryptographic proofs and DIDs to support privacy-preserving compliance at web-scale.

Complex financial contracts can be translated into executable smart contracts to support automation, auditability, and regulatory compliance. For example, AI4Contracts [58] demonstrates how large language models and retrieval-augmented generation can be used to encode over-the-counter derivatives contracts into smart contracts, enabling structured interpretation and enforcement of legally binding financial agreements in a gas efficient manner [56, 57]. While focused on contractual obligations rather than identity verification, this line of work highlights a shared objective: transforming regulated financial logic into machine-verifiable, decentralized execution. Our framework complements this direction by operationalizing compliance logic for identity verification and cross-border payments using cryptographic proofs and DIDs.

Furthermore, existing work shows that blockchain can serve as an effective trust layer for web-scale applications by supporting decentralized storage and AI-based verification in the context of misinformation [70] and even combating web spam [46]. We build on this principle in the CBPR setting by enforcing compliance through cryptographic proofs and smart contracts rather than centralized intermediaries.

Beyond data integrity and compliance enforcement, prior work has examined governance and safety mechanisms for smart contract-based systems. In particular, research on the feasibility of smart contract “kill switches” explores how controlled intervention points can be designed to halt or constrain contract execution under exceptional conditions, such as detected faults, attacks, or regulatory violations [71]. While such mechanisms introduce deliberate control into otherwise autonomous systems, they illustrate an important principle for real-world deployment: decentralized infrastructures must support auditable, policy-driven safeguards. Our framework similarly acknowledges this need by constraining authority to well-defined roles (e.g., CAs) and by enforcing compliance logic through verifiable smart contracts, rather than relying on unrestricted autonomy.

Together, these complementary threads underscore the growing role of blockchain and decentralized protocols in re-architecting web infrastructure toward verifiability, accountability, and user-centric control.

3 Stakeholders in Our zkKYC/zkCBPR Framework

The zkKYC/zkCBPR framework involves a set of clearly defined stakeholders, each playing a critical role in establishing trust, validating compliance, and ensuring seamless transactions. In the following subsections, we describe the main stakeholders and highlight their responsibilities within the protocol.

3.1 Users

Sender: This is the individual or institution initiating a payment. They generate proofs that they are KYC verified, possess sufficient funds, and comply with AML/CTF rules. Their DID anchors their identity, and proofs are shared with the receiver (via intermediaries) without exposing any of their personal data.

Receiver: This is the counterparty that ultimately accepts funds. Their role is to validate the sender’s proofs, often via their own bank or wallet provider. The receiver’s DID also links to VCs, ensuring both ends of the transaction are authenticated.

3.2 Participating Banks

Sender's Bank: This is the primary financial institution responsible for initiating a CBPR on behalf of the sender. It generates two artifacts before a payment is processed: (i) a ZKP for fund sufficiency, proving that the sender's account has enough balance to cover the transfer, and (ii) the transaction certificate, signed by the Intermediate Entity (IE), defined in Section 3.4, following successful certificate verification. These artifacts collectively establish that the transaction is authorized and compliant without exposing the sender's raw financial data. The sender's bank then submits both ZKPs to the next hop, either to an intermediary bank or the receiver's bank, along with the transaction payload.

Receiver's Bank: This is the final destination of the CBPR flow. Its primary function is to verify the validity of the sender's bank's proofs and the IE-issued transaction certificate. Once verification is complete, the receiver's bank credits the receiver's account with the specified amount. Additionally, it reports the transaction outcome back to the relevant IE, ensuring that the MTR, which is introduced in Section 4.2.2, is updated to reflect the completed payment in the user's transaction history.

Intermediary Banks: In many CBPR scenarios, funds pass through one or more intermediary banks before reaching the receiver's bank. Each intermediary bank performs ZKP and certificate verification at its hop, ensuring that the full payment chain remains secure and compliant. Upon successful verification, the intermediary bank forwards the payment and associated proofs to the next bank in the chain. This hop-by-hop verification process preserves trust while allowing settlement networks across multiple chains and regions to interoperate. As with the sender's and receiver's banks, the intermediary banks do not learn any raw user identifiers, as they only receive cryptographic proofs and certificates.

3.3 Certificate Authority (CA):

In the zkKYC framework, the CA is responsible for bootstrapping initial trust and ensuring secure credential issuance. The CA verifies users' real-world identities and issues DIDs, serving as a trust anchor for the system's initial credential binding. The CA is designed to leverage a widely trusted identity system to establish trust initially. For example, the CA will perform a one-time verification of the user's legal identity using government-issued documents (e.g., passports, national ID cards, social security numbers), and depending on policy requirements (i.e., identity tied to creditworthiness for a loan approval scenario), these identity checks may be corroborated with off-chain evidence such as credit history or proof of assets. All these real-world attributes presented by the user are verified within the CA's regulated and protected environment and are not embedded in the DID. After the verification, the CA issues VCs that tie the identity of a user-controlled DID.

Importantly, the CA's responsibilities are limited to initial issuance: once a DID is generated and registered on-chain, credential verification no longer depends on the CA. All subsequent verifications are trust-minimized, conducted through smart contracts and ZKPs, which enable verifiers to validate user credentials without relying on any central authority or accessing underlying personal data. This design ensures that identity proofs are both privacy-preserving and peer-verifiable in a decentralized manner. In fact, the CA functions as the root of trust, issuing root certificates to authorized IEs. Each IE, in turn, signs the "transaction certificates," which is introduced in Section 4.3.2, for users whose credentials and transaction requests have been verified. Anyone can validate this chain step by step: beginning with the CA's root certificate, confirming the authenticity of the IE's certificate, and finally verifying the sender's "transaction certificate." This layered validation ensures that every entity in the chain, from the root authority to the individual user, has been properly vetted and cryptographically endorsed.

To minimize reliance on any single entity that could lead to undesired centralization, the CA role can be implemented as a decentralized or federated consortium of trusted issuers (e.g., banks, government agencies, or accredited institutions). Such a model mitigates the risk of a single point of failure and aligns with the DID

governance principles adopted in existing SSI ecosystems. While the CA serves as an initial trust anchor, our framework is explicitly designed to minimize long-term reliance on any single authority. Several mitigation strategies are possible if a CA is compromised or behaves maliciously. First, the CA role can be instantiated as a federated or consortium-based authority, requiring threshold agreement among multiple independent issuers before a DID or credential is accepted. Second, all CA-issued artifacts are publicly auditable on-chain, enabling post hoc detection of mis-issuance or policy violations. Finally, compromised CAs can be revoked through governance mechanisms already present in existing SSI ecosystems, such as trust registries or revocation lists, without invalidating previously verified transactions.

3.4 Intermediate Entity (IE):

The IE plays a semi-trusted but privacy-preserving role in the framework, acting as a validator of transaction-specific information without ever accessing raw personal data. IEs are particularly critical in CBPR scenarios where the sender and receiver are rarely directly connected. For example, a sender may bank with a small regional bank in one country while the receiver holds an account at a regional bank in another country. Since these two banks often lack a direct payment corridor, IEs are required to bridge the gap, validate compliance obligations, and route the transaction across multiple jurisdictions. IEs provide this connective tissue in a decentralized way, ensuring that compliance checks and proof validations occur even when participants are separated by geography and financial networks.

When initiating a transaction, a user (i.e., a sender), submits a Certificate Signing Request (CSR) to the IE. The primary purpose of a CSR is to obtain a cryptographic attestation that the user's identity, credential status, and transaction details have been verified. The IE verifies the integrity of the request by checking the user's DID registration on-chain, confirming that wallet addresses and transaction details are consistent. After these checks, the IE issues a signed certificate that serves as a trust endorsement for subsequent ZKPs.

Importantly, IEs are not single points of control: multiple IEs may operate in parallel, with their public keys registered on-chain and their behavior enforced through smart contracts. This decentralized IE model allows banks, payment processors, or accredited industry consortia to collectively serve as validators, reducing systemic risk while supporting interoperability across different blockchains and payment networks. In essence, the IE functions as a bridge between user-controlled identities and the wider dApp ecosystem, enabling compliance validation, cross-chain credential recognition, and trust minimization without compromising user privacy.

4 Methodology

This section details the design and implementation of the zkKYC/zkCBPR framework, outlining the steps required to establish a DID, generate privacy-preserving ZKPs, and enable secure CBPR. Together, these components form a cohesive protocol that eliminates reliance on centralized intermediaries and supports verifiable compliance. In Figure 1, we provide a high-level overview of our system architecture, which introduces a comprehensive two-phase architecture based on two core components: zkKYC/zkCBPR.

4.1 Generation of the Unique DID

Figure 2 illustrates the process for DID generation and registration. During this phase, users establish a verifiable, self-controlled digital identity without revealing any sensitive personal information. The process begins with the CA verifying the user's real-world identity, then generating a unique DID and registering it on the blockchain. This initial setup ensures that every participant in the protocol has a legitimate real-world identity and is compliant with regulatory standards from the outset.

Specifically, after verifying the user's real-world identity, the CA generates a cryptographic hash value using the SHA-256 hash algorithm that combines the user's name, ID, and other information to form a unique fingerprint

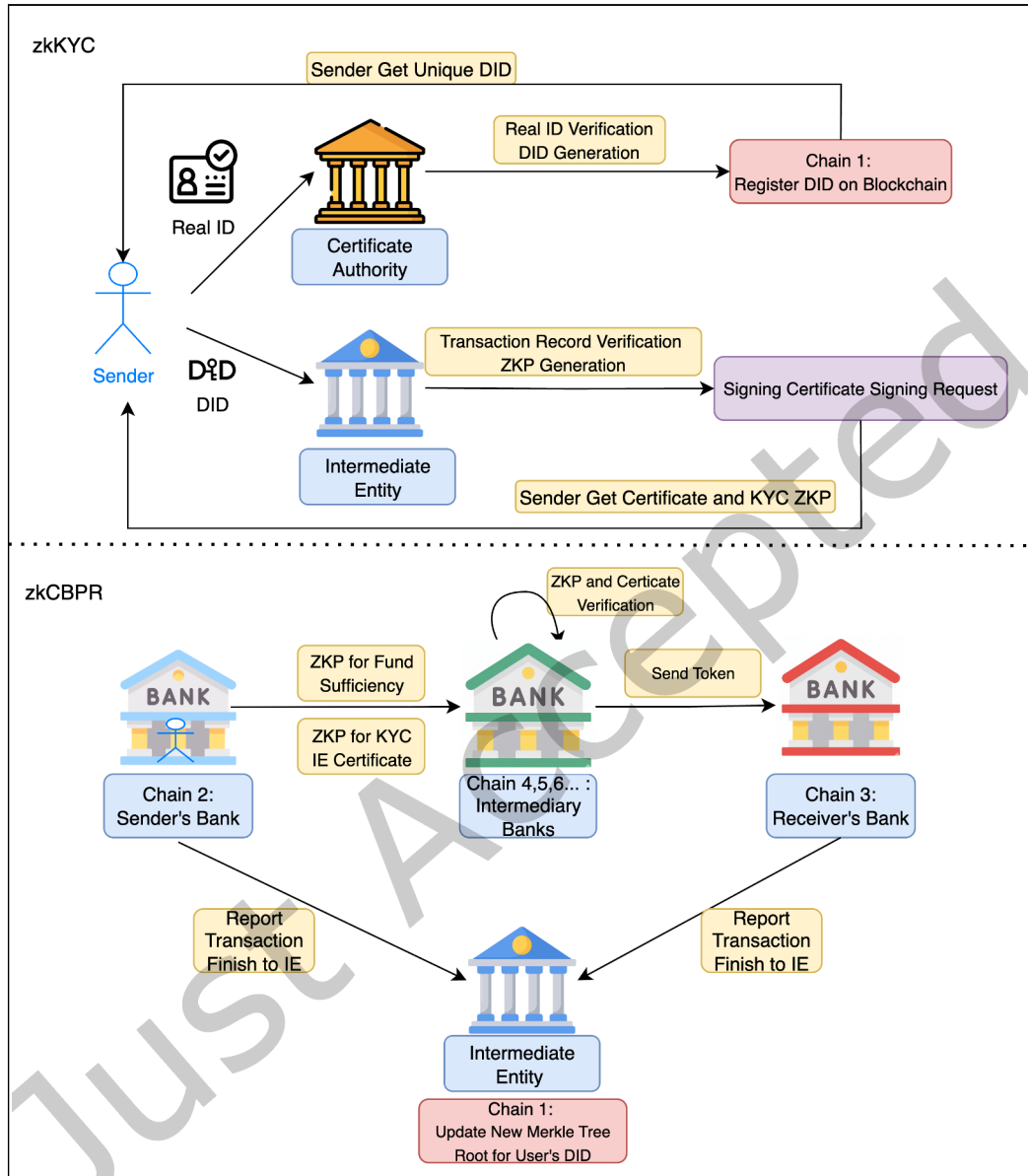


Fig. 1. The end-to-end zkKYC/zkCBPR workflow implementing CBPR with DID verification.

of the user's identity. The generated hash is then used to construct a DID in the format of `did:kyc:hash`, which the CA signs with its private key. This digital signature proves the authenticity of the DID and associates it with the CA. Once signed, the DID is registered on the blockchain by the CA through the `MerkleTreeWithDID` smart

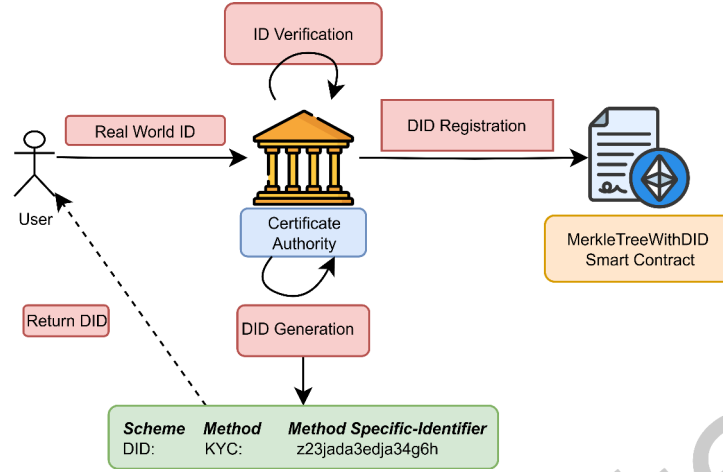


Fig. 2. DID Generation and Registration.

contract, making it publicly verifiable while ensuring that the underlying identity information remains secure and private.

4.2 zkKYC Architecture

The zkKYC component is the first phase of our two-phase design (zkKYC $\rightarrow$ zkCBPR). Its purpose is to establish a verifiable, privacy-preserving identity and compliance state for a user *before* any payment is attempted, and consists of several steps that are outlined below.

4.2.1 Certificate Signing Request (CSR). A CSR is a formal request submitted by a user to an IE to verify and authenticate a transaction. The CSR contains essential information, including the user's DID, transaction details, and other relevant data crucial to the verification process. For example, consider a CBPR scenario where a small business in Country A is sending funds to a supplier in Country B. Before the payment is executed, the sender's bank must ensure that the business is KYC-verified, has no outstanding compliance flags, and possesses sufficient funds to complete the transfer. The business submits a CSR containing its DID, the payment amount, and relevant transaction details. This request is sent to the IE, which serves as a neutral validator. Upon receiving the CSR, the IE first verifies its authenticity. This involves a series of security checks, such as verifying whether the DID provided in the CSR has been correctly registered on the blockchain, confirming that the wallet address associated with the DID matches the submitted transaction information, and ensuring the transaction history is valid. Once the information in the CSR is confirmed to be accurate, the IE signs it, converting it into a formal certificate. By signing the CSR, the IE offers a strong trust endorsement for the transaction and its associated information. The signed certificate proves the transaction's authenticity and ensures the user can participate in future zkKYC transactions. In such transactions, the signature certificate is used to verify the identity of users and the validity of the transaction without revealing sensitive personal information, such as real names or identification numbers. Figure 3 illustrates this process in the context of CBPR: the CA first certifies the IE, the IE validates the user's CSR, and the signed certificate can then be verified by banks using the CA's and IE's public keys. This chain of signatures ensures that the user's transaction request is cryptographically trustworthy before the payment is executed.

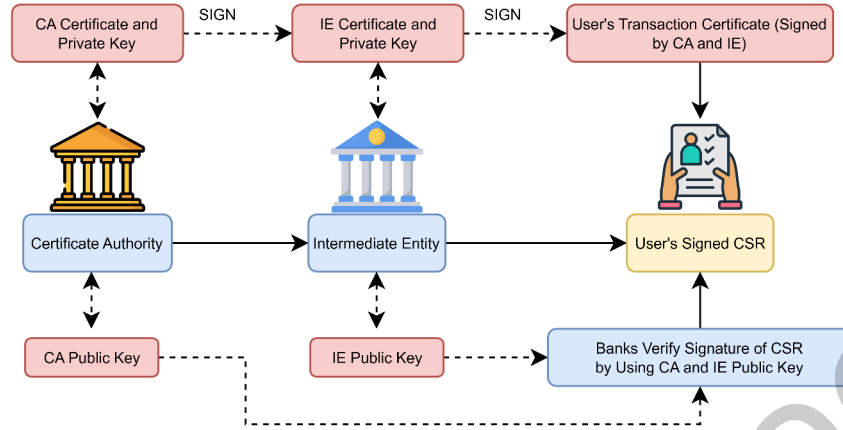


Fig. 3. Process of Certificate Signing Request (CSR) signing and verification.

4.2.2 Ensuring Transaction Record Integrity with Merkle Tree Root (MTR) and ZKP. Once the CSR is verified by the IE, the user's transaction history must be returned in a manner that preserves both privacy and trust, and our framework employs a combination of Merkle tree structures via the MTR and ZKPs for this purpose.

The process begins with creating a Merkle tree using the user's transaction history. Each individual transaction is represented as a leaf node in the Merkle tree as illustrated in Figure 4. These leaf nodes are then repeatedly combined and hashed in pairs until a single value, the MTR, is produced. Any change to an individual transaction record would result in a completely different MTR, making it a reliable indicator of transaction history integrity.

In our framework, the MTR generated in the ZKP circuit must be consistent with the one on the blockchain. In this circuit, we provide the MTR as a public input, along with a specific leaf node (representing a transaction). The IE then generates a proof demonstrating that it knows the path from that transaction (the provided leaf node) to the MTR. Successfully generating this proof shows two things: (1) the IE indeed has access to the full transaction history, and (2) the transactions have not been altered, since the path and root match exactly.

All proof generation during this process is private and secure, because the zkKYC framework utilizes ZKP circuits that run entirely within IE's local environment, avoiding exposing raw personal data to any external verifier. Sensitive inputs, such as real-world identity attributes, transaction history, or wallet linkage information, are first cryptographically committed to Merkle trees before being passed into the circuit. This ensures that only the resulting ZKPs, not the underlying data, are ever published on-chain. Secure hashing algorithms (e.g., SHA-256) preserve the integrity of the input data, while non-interactive zkSNARK proofs [11] provide soundness and zero-knowledge guarantees, protecting against replay attacks and data interception during transmission.

Furthermore, the private keys used to sign DIDs, CSRs, and transaction certificates are always stored on secure hardware reducing the possibility of key compromise. Verifiers receive only concise cryptographic proofs that can be publicly verified directly against on-chain commitments without revealing, storing, or transmitting the underlying personal data. Additionally, we specifically leveraged the MTR as an off-chain verification mechanism, as it is impractical to store everyone's entire transaction history in the full Merkle tree on-chain, given its size and complexity. We only store the MTR and the corresponding user's DID on the blockchain.

4.3 zkCBPR Architecture

After completing the zkKYC process, the user has obtained a signed KYC certificate and an MTR-based ZKP. The zkCBPR phase handles the execution of actual financial transactions, building on the trusted identity and compliance proofs established during zkKYC. When the sender initiates a payment through their bank, the bank generates a ZKP proving that the user has sufficient funds to cover the transaction. The intermediary bank then receives three key artifacts: the transaction certificate, a ZKP attesting to the integrity of the sender's historical transaction record, and a ZKP confirming the sufficiency of funds. Using these inputs, the intermediary bank can efficiently verify both the authenticity of the certificate and the validity of the proofs.

Once verification is successful, the intermediary bank transfers the payment amount to the next bank in the chain. This process repeats until the receiver's bank receives the final transfer. After settlement, both the sender's bank and receiver's bank submit a full report of the transaction to an IE. This final step updates the MTR associated with the DID, ensuring that the transaction history remains complete, tamper-evident, and ready for generating proofs in subsequent transactions.

4.3.1 ZKP for Bank Signature and Fund Verification. In the zkCBPR framework, the *sender's bank* is crucial for verifying the sufficiency of funds and generating a ZKP to validate the transaction while maintaining the user's privacy. The process begins with the *sender's bank* generating a pair of public key coordinates A_x , A_y , and private

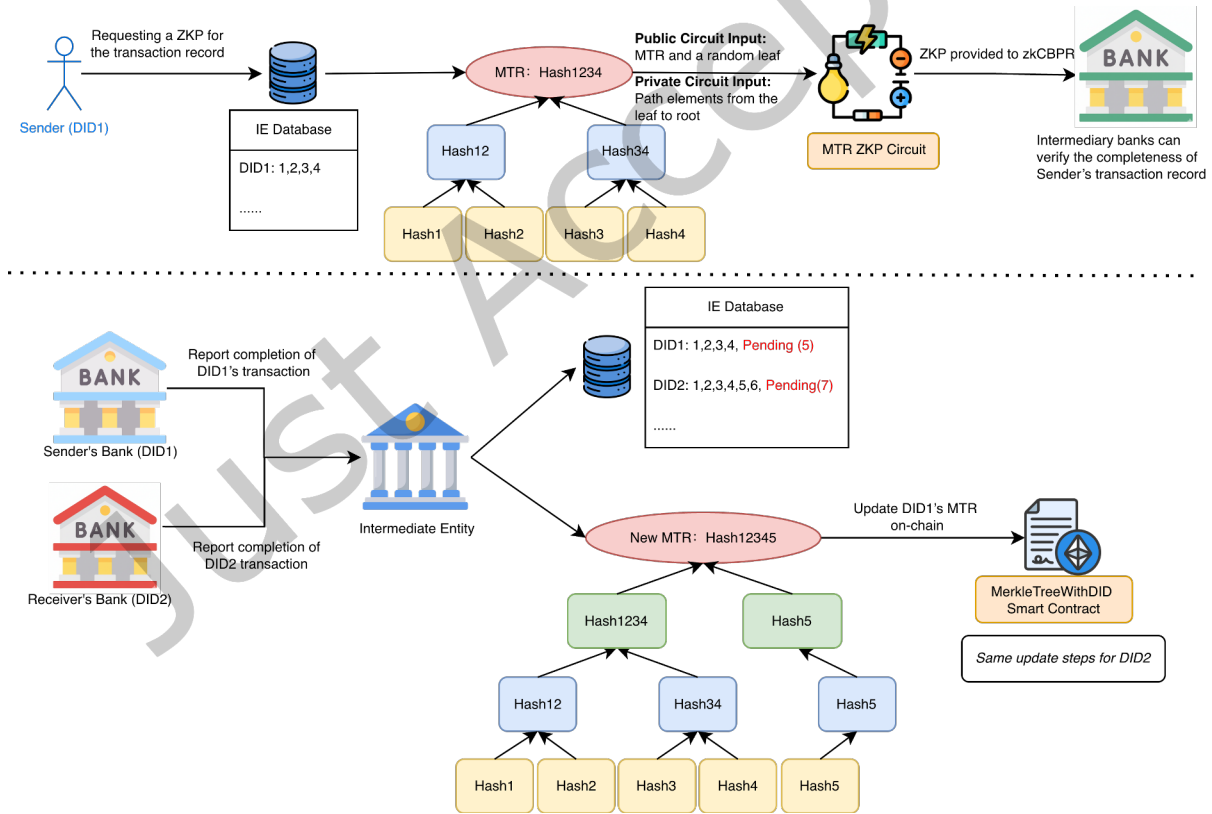


Fig. 4. Merkle Tree-Based Structure for zkCBPR.

key R_{8x} , R_{8y} using the Edwards-curve Digital Signature Algorithm (EdDSA) [45]. The private key is used to sign the transaction details, while the corresponding public key is used for identity verification. The ZKP circuit takes the transaction amount and public key as public inputs, while the sender's total assets and the bank's private key are provided as private inputs. The goal of the circuit is to verify whether the sender has sufficient funds in their *sender's bank* account to cover the transaction. The circuit ensures that the total assets exceed the transaction amount. After verifying the sufficiency of funds, the *sender's bank* signs the transaction using their private key. The proof generated by the circuit serves as a trusted confirmation that the sender has enough funds to complete the transaction. Although the *sender's bank* will send the required funds to the *intermediary bank* via a smart contract (CBPRPaymentSystem smart contract), this proof is necessary for compliance and record-keeping. If the sender is subject to future audits, the *sender's bank* can provide its original input file to facilitate any regulatory oversight.

4.3.2 Verification and Settlement. Before any transaction, the sender must hold: (1) the transaction details, (2) the MTR ZKP, and (3) the "sufficient funds" ZKP. The *intermediary bank* then takes on the task of verifying the proofs and facilitating the transfer from the CBPRPaymentSystem smart contract to the *receiver's bank* or the next target. The *intermediary bank* or the *receiver's bank* (in the case of a direct transfer between the sender and receiver) first verifies the integrity and authenticity of the transaction using the two ZKPs. The MTR ZKP focuses on verifying the completeness of the transaction record represented by the MTR. The "sufficient funds" ZKP checks whether the sender's signature is valid and the sender has sufficient funds. Both ZKPs are verified with our implementation developed using the Snarkjs library [44].

During verification, the proofs are checked against a set of *public inputs*, which are values that are made available to any verifier. These typically include the MTR, the transaction amount, and the sender's and receiver's DIDs. Public inputs act as anchors: they allow anyone to confirm that the ZKP was generated for the correct transaction parameters without learning any private information (e.g., the sender's exact balance or the full transaction history).

After verifying the ZKPs, the *intermediary bank* proceeds with "transaction certificate" validation, a step centered around constructing and verifying the certificate trust chain. In this context, the "transaction certificate" is a signed artifact issued by the IE after verifying the sender's DID, credential status, and transaction details via the CSR process. It serves as a cryptographic attestation that the transaction request has passed compliance checks (e.g., KYC/AML verification) and is authorized to proceed. This "transaction certificate," combined with the two ZKPs, allows the intermediary bank to validate both the legitimacy of the sender and the correctness of the transaction state without accessing sensitive user data. Subsequently, the *intermediary bank* performs certificate validation by reconstructing and checking the entire trust chain.

Once all verifications have been completed, the *intermediary bank* proceeds with the fund transfer. The funds initially received from the *sender's bank* through the CBPRPaymentSystem smart contract can now be transferred to the receiver or the next target bank in the transaction chain.

4.3.3 Final Transaction Recording and MTR Update. The transaction process is complete once the *receiver's bank* has received the funds. At this stage, both the *receiver's bank* and the *sender's bank* must send the transaction details to the IEs in the transaction chain. Each IE first validates the received transaction details, then appends the new record, containing the sender's and receiver's DIDs, transaction amount, timestamp, and the participating banks or IEs, to its locally maintained transaction history. This step ensures that the IE's off-chain ledger remains synchronized with the most recent settlement.

Once the local history is updated, the IE recalculates the MTR by incorporating the newly added transaction as a leaf node and recomputing all necessary intermediate hashes. After recording the transaction, the IEs publish the newly computed MTR to the blockchain using the MerkleTreeWithDID smart contract. This on-chain update

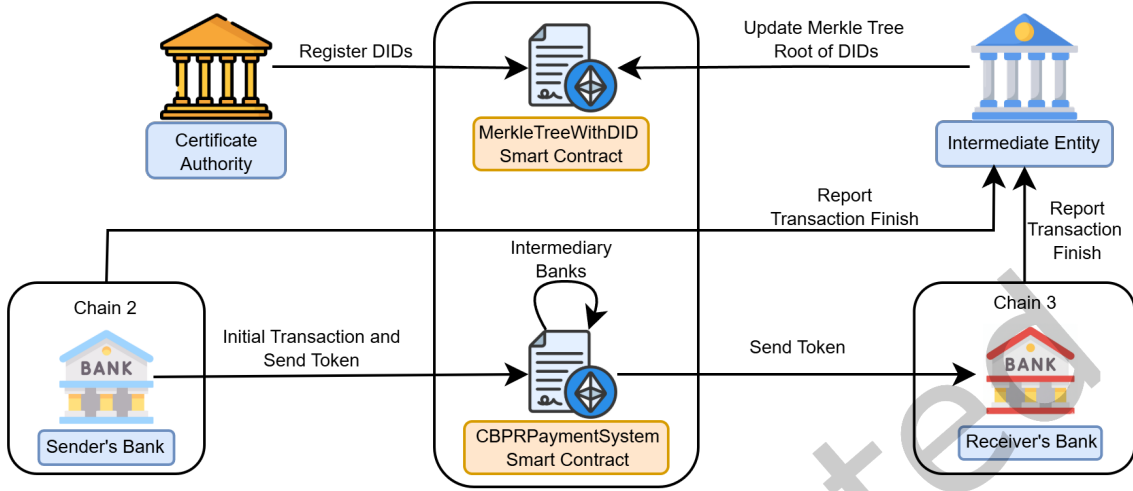


Fig. 5. Overview of the zkKYC/zkCBPR Smart Contract Logic.

acts as a public commitment to the new state of the transaction history, allowing any verifier to later check proofs against the most recent root.

The MTR functions as a cryptographic fingerprint of the complete transaction record. By publishing it after every new transaction, the IE ensures that the ledger remains tamper-evident, auditable, and ready for generating ZKPs for future transactions.

4.4 Smart Contracts

The core functionality of the zkKYC/zkCBPR framework is encapsulated within two primary smart contracts. The `MerkleTreeWithDID` contract manages the DIDs, while the `CBPRPaymentSystem` contract facilitates secure and transparent CBPR transactions. An overview of their logic is shown in Figure 5, with detailed execution illustrated in the sequence diagrams Figures 6 and 7. Steps in processes such as credential issuance and payment verification are explicitly numbered (e.g., 2.1, 2.2, 2.3) to clarify the logical flow and dependencies within and between the smart contracts.

4.4.1 MerkleTreeWithDID Smart Contract. This smart contract manages DIDs and maintains their associated cryptographic proofs using MTRs. Key functions include:

- **DID Management and Registration:** Ensures secure registration and uniqueness of user identities by restricting DID issuance authority exclusively by known CAs. The selected CA verifies user attributes off-chain, then records the DID securely within the contract.
- **Merkle Tree Integration:** Integrates with Merkle trees to guarantee data integrity and immutability of credential updates associated with DIDs. After each credential update or new transaction, authorized IEs update the MTR stored on-chain, thus preserving a cryptographically verifiable audit trail.
- **Authorized Entity Updates:** Only authorized IEs, authenticated by the CA, can update the MTR within the contract, enforcing a controlled yet decentralized mechanism for maintaining data integrity and security.

The deployed `MerkleTreeWithDID` contract can be accessed on the Ethereum Sepolia Testnet at: <https://sepolia.etherscan.io/tx/0x0cb66ca4b1430d0491ddb3a17de7298bb23fdeb3070c3c17383d48dd00fe960a>.

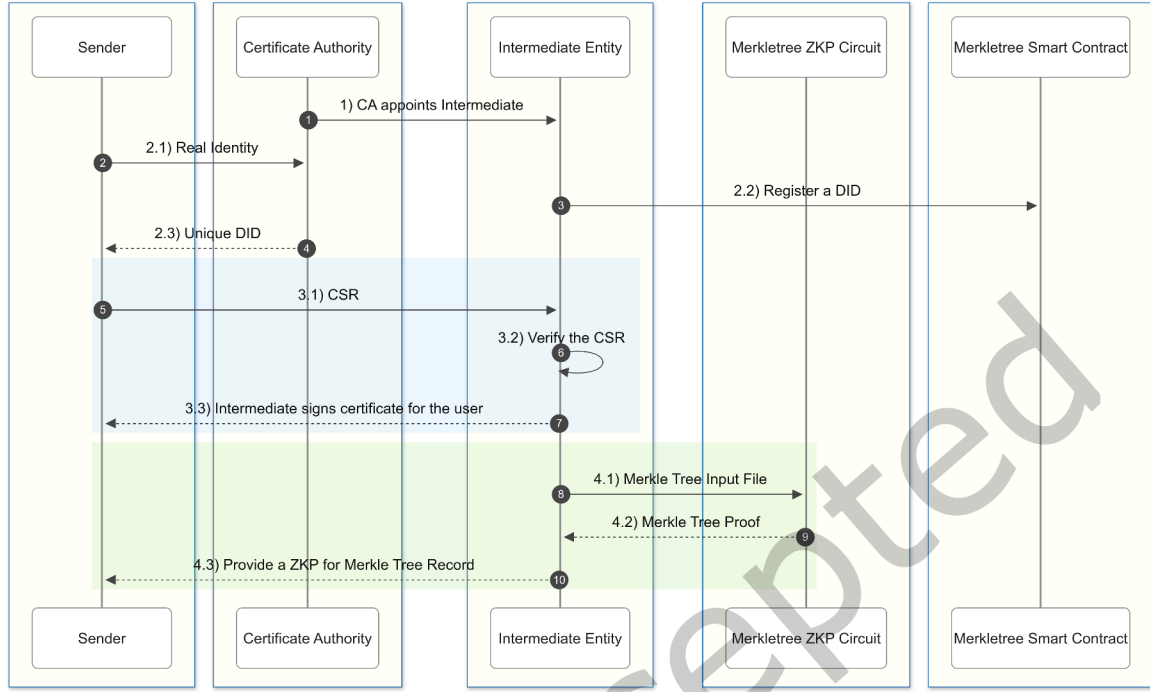


Fig. 6. zkKYC Workflow with the MerkleTreeRootDID Smart Contract

4.4.2 CBPRPaymentSystem Smart Contract. This contract is central to managing the overall CBPR processes. It securely handles funds transfers and maintains immutable records, enhancing both trust and compliance across payment networks. Key functionalities include:

- **Secure Fund Transfers:** Facilitates direct blockchain-based transfers of funds and accompanying messages to the receiving bank's smart contract address.
- **Immutable Transaction Records:** Records each transaction event directly onto the blockchain, leveraging Ethereum's built-in immutability and transparency. Each payment triggers verifiable confirmation events.

The deployed CBPRPaymentSystem contract can be accessed on the Ethereum Sepolia Testnet at: <https://sepolia.etherscan.io/tx/0x7cbc5789048a62d68ec784a25c1c3bd7124db12071234d3474d59f5b295bdf3c>.

4.4.3 Deployment Environments. To ensure reliable development and testing, we utilized both local and test network environments, as described below.

- (1) **Local Development Environment:** We utilized Anvil [23], a local Ethereum development tool from the Foundry suite. Anvil provides an efficient blockchain emulation environment tailored for rapid iteration, smart contract deployment, and thorough debugging.
- (2) **Ethereum Sepolia Test Network:** For a more realistic evaluation scenario, we deployed our smart contracts on the Sepolia Testnet [20], which offers a controlled blockchain environment closely mimicking Ethereum's main network.

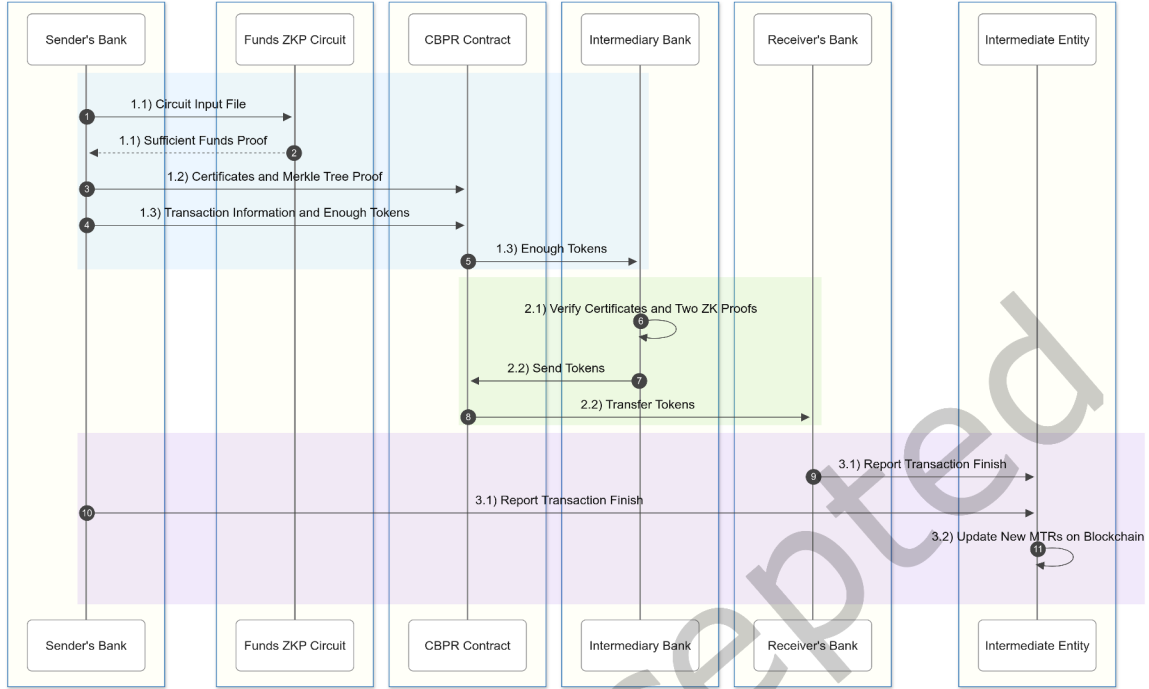


Fig. 7. zkCBPR Workflow with the CBPRPaymentSystem Smart Contract

5 Evaluation

Our analysis focused on key features critical to CBPR, comparing our framework against mainstream solutions, and the efficiency of the framework by examining key performance metrics such as ZKP generation and verification time, transaction time, and transaction fees.

5.1 Comparative Evaluation Against Existing DID Solutions

To contextualize our zkKYC/zkCBPR framework within the broader DID ecosystem, we compare against several prominent research prototypes and commercial solutions as shown in Table 1. These platforms represent diverse approaches to SSI, ranging from selective disclosure systems to zero-knowledge-native architectures and enterprise-focused compliance solutions. Our evaluation considers key metrics relevant to web-scale identifier systems, such as DID and VC support, privacy via ZKPs, cross-chain interoperability, alignment with W3C standards, smart contract integration, and regulatory reporting capabilities.

The results in Table 1 highlight that our zkKYC/zkCBPR framework is the only evaluated solution that simultaneously provides: (i) W3C-conformant DID and VC support, (ii) ZKP-based privacy-preserving verification for both KYC and transaction compliance, and (iii) built-in audit-friendly regulatory reporting. While Quadrata [35] and Verite [16] offer strong DID/VC implementations, they lack native ZKP privacy guarantees, relying instead on attestations. Zeeve ID [95] and Holonym [34] both leverage ZKPs but focus primarily on user privacy or ZK-native infrastructure, offering limited support for CBPR-style reporting or institutional audit requirements. In addition, our framework distinguishes itself by supporting federated trust minimization through a CA-IE hierarchy, enabling multiple IEs to operate in parallel and reducing reliance on single points of control. This

Table 1. Comparison of Our zkKYC/zkCBPR with leading DID frameworks.

	zkKYC/zkCBPR (Our Framework)	Quadrata [35]	Verite [16]	Zeeve ID / Privado [95]	Holonym [34]
<i>DID Support</i>	Yes – uses W3C-conformant DIDs registered on-chain	Yes – off-chain verified, then on-chain NFT-based identity	Yes – uses W3C-compliant DIDs and VCs	Yes – supports iden3 DID, integrated with rollup infra	Yes – uses zkID based on privacy-preserving DID formats
<i>Zero-Knowledge Privacy</i>	Yes – ZKPs used to verify KYC and compliance attributes without revealing personal data	No – no ZKPs, but offers verifiable attestations	No – relies on off-chain JWT/VC models	Yes – ZKPs core to ZKP generation and verification	Yes – ZKPs used for identity, residency, and human verification
<i>Regulatory Reporting Support</i>	Yes – designed for AML/CTF, CBPR compliance with audit-friendly proof logging	Yes – supports regulatory-grade KYC attestation tokens	Yes – flexible compliance via credential frameworks	Yes – tailored to ZK-compliant DeFi/KYC integration	Partial – focused on user privacy; not audit-logging oriented
<i>Cross-Chain Compatibility</i>	Yes – VCs and ZKPs can be used across blockchains	Yes – works across EVM-compatible chains	Yes – blockchain-agnostic VC layer	Yes – compatible with EVM chains and rollups	Yes – credentials verifiable on multiple EVM chains
<i>Web Standards Alignment</i>	Yes – conforms to W3C DID and VC standards	Yes – uses standards plus internal token formats	Yes – fully aligned with W3C DID/VC specs	Yes – follows SSI with ZK claim formats	Partial – not W3C-native but aligns with DID principles
<i>Use of VCs</i>	Yes – ZK-verifiable VCs issued and held by users off-chain	Yes – attestation NFT or passport-based VCs	Yes – reusable, off-chain VCs	Yes – ZK-compatible credentials with iden3 format	Yes – zkID proofs include VCs for human and location attestations
<i>Trust Model</i>	Federated CA issues credentials; trust minimized via ZKPs and on-chain proofs	Off-chain verification by trusted partners, tokenized on-chain	Issuers define trust policies and schemas	Decentralized trust via identity trees and contracts	Federated attestation model with ZK-based trust minimization
<i>Smart Contract Integration</i>	Yes – DID registration, Merkle updates, and ZKP verification handled on-chain	Yes – identity NFTs and attestation checks on EVM chains	No – primarily off-chain credential verification	Yes – EVM smart contracts handle proof and VC logic	Yes – on-chain verification of zkID proofs using Solidity libraries
<i>Deployment Status</i>	Prototype – fully implemented and evaluated in testnet	Live – integrated in DeFi platforms, such as Ondo and Masa)	Prototype framework-level, not a single platform	Live – deployed rollup ID framework with zk toolkits	Live – zkID available, zk residency & KYC flows testable on Polygon

design allows for scalable cross-chain interoperability, with MTRs and ZKP verification executed directly on-chain. Together, these characteristics position zkKYC/zkCBPR as a bridge between compliance-oriented institutional requirements and privacy-preserving, decentralized identity systems, offering a deployment-ready approach suitable for web-scale CBPR ecosystems.

5.2 Comparative Evaluation Against Existing CBPR Solutions

Next, we systematically evaluated the performance and characteristics of various CBPR solutions, focusing on key metrics such as privacy protection, transaction efficiency, decentralization, regulatory compliance, cost, and transparency. Table 2 defines the evaluation metrics that provide a uniform basis for comparison of various commercial solutions such as Swift Global Payments Innovation (GPI) [82] and Visa B2B Connect [7, 88]. All

analyses and insights were derived from the official transaction process descriptions on the respective platforms' official documentation.

As seen in Table 3, our zkKYC/zkCBPR framework provides additional contributions in several key areas compared to existing CBPR systems. Most notably, our zkKYC/zkCBPR framework emphasizes stronger privacy protection by ensuring no personal information is exposed through ZKP. Furthermore, our solution offers a higher degree of decentralization, reducing reliance on centralized intermediaries and providing a more distributed control model. Finally, transparency is also a central focus of our framework, with blockchain technology ensuring full transparency and auditability of all transactions.

5.3 ZKP generation and Verification Time

Before presenting the empirical results, we briefly clarify the performance characteristics of the ZKP components used in zkKYC/zkCBPR. As is typical for zkSNARK-based systems, proof generation is the most computationally intensive step, as it requires witness construction and circuit execution over private inputs. In contrast, proof verification is lightweight and incurs nearly constant time overhead independent of transaction history size. This asymmetry provides a powerful advantage in our application: expensive computation is performed locally by trusted parties (e.g., banks or IEs), while verifiers incur minimal cost, enabling scalable and low-latency validation

Table 2. CBPR Platform Evaluation Metrics Definitions

	High	Moderate	Low
Privacy Protection	No personal data exposed to third parties.	Data is protected by encryption and accessible only to authorized entities.	Data shared with intermediaries, exposing privacy risks.
Verification Time and Transaction Efficiency	Fully automated, completed in minutes.	Faster than traditional, some manual steps are required.	Slow, manual checks or IE delays.
Decentralization	Fully decentralized, no central authority.	Partially decentralized, key control retained by entities.	Centralized, controlled by few entities.
Regulatory Compliance	Fully compliant with all regulations.	Mostly compliant, third-party involvement for some tasks.	Partially compliant or lacking key regulatory features.
Cost Transparency	All fees transparent and predictable.	Most fees are disclosed with some hidden costs.	Fees unclear and hidden charges are common.
Time Transparency	Full visibility and real-time tracking of transactions.	Partial visibility, estimated completion times not guaranteed.	Little to no transparency on transaction duration.
Transaction Transparency	Fully visible and traceable transactions.	Partial visibility controlled by intermediaries.	Opaque transactions, minimal visibility for users.

at web-scale. We evaluated the performance and cost-effectiveness of our framework by analyzing key metrics as demonstrated in Figures 8 and 9, and Table 4. We conducted all the following experiments on a MacBook Pro equipped with an Apple M3 Pro chip (12-core CPU, 18-core GPU) and 36 GB of memory.

First, we emphasize the difference between MTR proof generation and zkMTR proof generation presented in Figure 8. MTR proof generation refers to the deterministic computation of the MTR by hashing all transaction records and combining them into a single cryptographic fingerprint. This step is computationally lightweight and

Table 3. CBPR Solution Comparison

	zkKYC/zkCBPR (Our Framework)	Swift GPI [82]	Visa B2B Connect [88]
Privacy Protection	High, ZKP ensures no personal information exposure.	Low, personal information is needed in ISO 20022 standard [79] and will be transferred to the <i>intermediary banks</i> .	Moderate, personal information is required but protected through Visa's secure network [87].
Verification Time and Transaction Efficiency	High verification, ZKP generation, and payment completed in a few minutes.	Moderate, fast payment speed, 90% of payment could be finished in 24 hours. But unknown verification time for each transaction [80].	Moderate, much faster than traditional banking systems but dependent on Visa's network [90].
Decentralization	Moderate, partially decentralized but subject to regulatory oversight.	Low, centralized under Swift's control with participating banks [82].	Low, centralized, with control under Visa and participating financial institutions [86].
Regulatory Compliance	High, fully compliant by CA and IE.	High, compliant with global regulatory standards [81].	High, compliant with international financial regulations [92].
Cost and Cost Transparency	High, transparent blockchain gas and low transaction fees.	Moderate, transparent fees but cost depends on service fees, like different <i>intermediary banks</i> [78].	Low, no intermediary fees but Visa B2B connect service fees may apply [91].
Time Transparency	High, transaction could be finished in a few seconds on blockchain.	Moderate, some stages are transparent, but exact completion times may vary [78].	Moderate, most transaction phases are visible and on time but exact timings are not clear [89].
Transaction Transparency	High, fully visible and auditable blockchain records.	Low, visibility controlled by banks and intermediaries.	Low, visibility controlled by financial institutions on Visa's network.

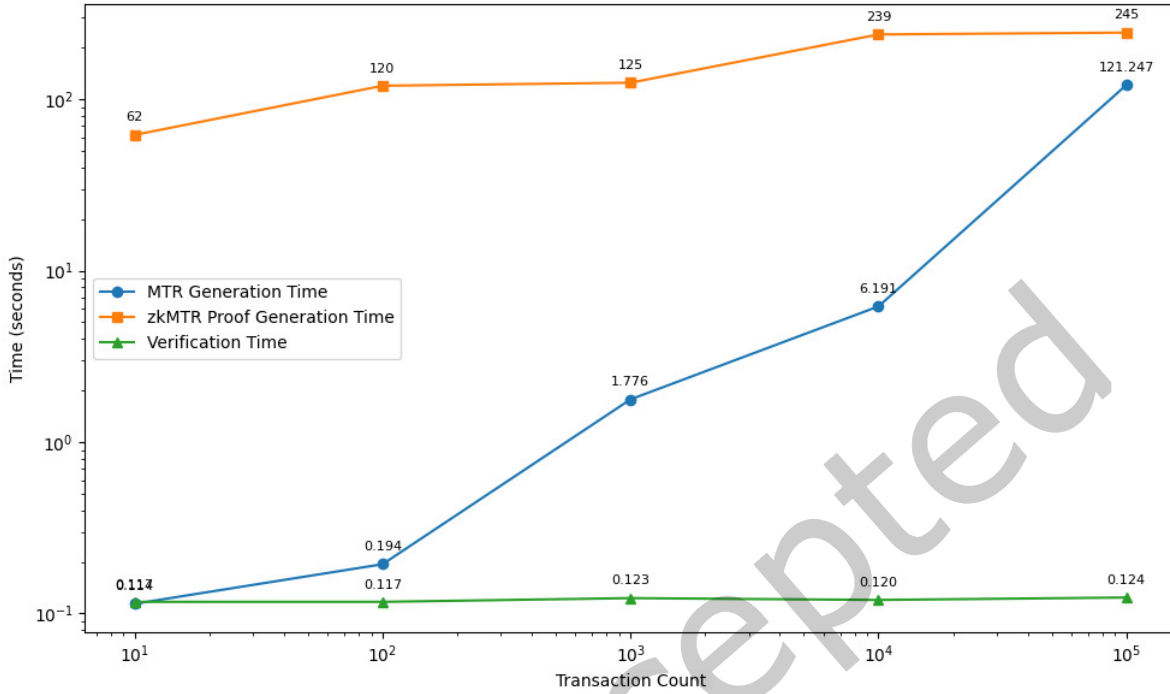


Fig. 8. zkKYC/zkCBPR MTR ZKP Generation and Verification Times

scales linearly with the number of transactions. In contrast, zkMTR proof generation involves constructing a zero-knowledge circuit that proves knowledge of the correct path from a given transaction leaf to the computed MTR without revealing the underlying transaction data. This process is significantly more computationally intensive, as it requires generating cryptographic witnesses and producing a zkSNARK proof. While MTR generation ensures data integrity, zkMTR proof generation provides privacy-preserving verifiability, allowing external parties to trust the result without directly accessing the transaction history.

We evaluated the performance of our zkKYC/zkCBPR framework by measuring both the MTR and zkMTR proof generation and the proof verification times under increasing transaction volumes. Figure 8 shows the performance for generating and verifying proofs based on different sizes of transaction amounts averaged from five independent runs. The MTR proof generation time increases gradually as the transaction volume grows. Smaller transaction counts (10–100 records) remain minimal, ranging from 0.114 to 0.194 seconds. However, even when the transaction count reaches 10^5 , the MTR proof generation time only rises to 121.247 seconds.

Similarly, the zkMTR proof generation times maintain excellent efficiency as the number of transactions increases. For transactions up to 100, the zkMTR proof generation time remains stable between 62 and 120 seconds. Even as the number of transactions increases to 10^5 , the zkMTR proof generation time only extends to 245 seconds. This confirms that the computational overhead introduced by ZK circuits remains manageable even for large datasets.

Importantly, verification time for both types of proofs remains almost constant across all transaction volumes, ranging between 0.117–0.124 seconds. This flat profile demonstrates that our approach can sustain low-latency verification regardless of the transaction history size, enabling near real-time validation at web-scale.

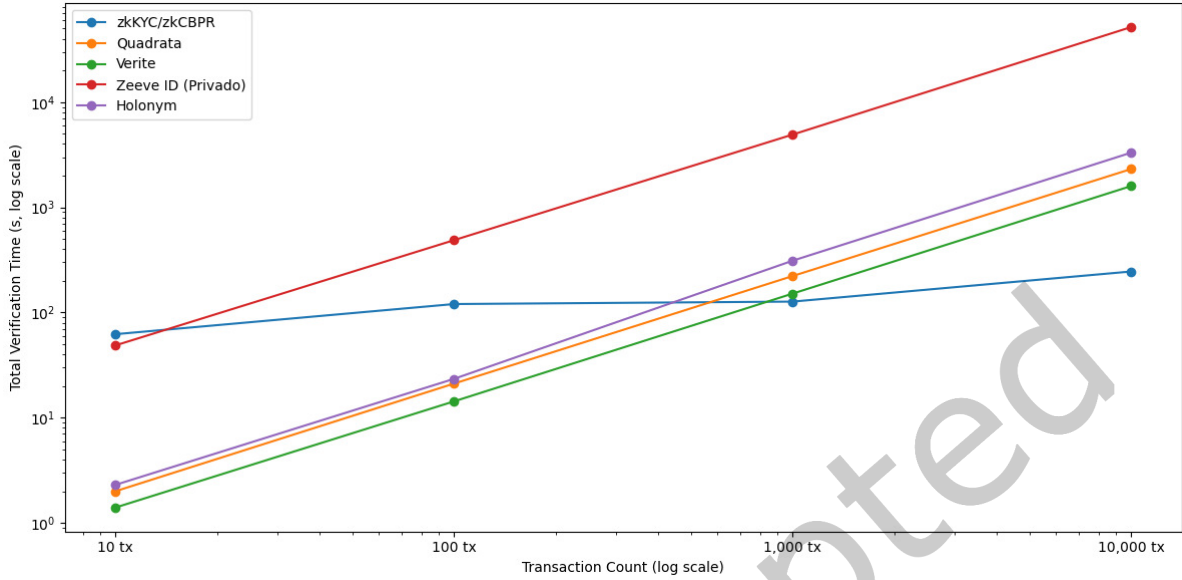


Fig. 9. Verification Time Comparison Across Frameworks

Additionally, to evaluate scalability relative to existing DID solutions, we compared total verification times across frameworks, as shown in Table 4 and Figure 9. While zkKYC/zkCBPR exhibits higher constant overhead for very small transaction volumes (10–100 transactions) due to initial circuit setup and witness generation, it significantly outperforms competitors at larger scales. For 10^4 transactions, zkKYC/zkCBPR completes verification in 366.4 seconds, which is over $6\times$ faster than Quadrata, $9\times$ faster than Holonym, and more than $140\times$ faster than Zeeve ID. This scalability advantage stems from our use of a Merkle tree structure to aggregate transaction history, enabling logarithmic path proofs rather than per-transaction linear checks. Competing solutions that verify transactions individually incur near-linear or super-linear growth in runtime as dataset size increases. Our design amortizes ZKP generation cost across the entire transaction set, while keeping verification latency nearly constant, making zkKYC/zkCBPR highly suitable for high-throughput, compliance-heavy CBPR scenarios. This design choice not only reduces computational overhead, but also ensures that verification latency remains nearly constant even as transaction count increases, making zkKYC/zkCBPR especially well-suited for high-throughput, scalable identity verification systems.

Table 4. Total Verification Time Comparison Across Frameworks

	10 tx	100 tx	1000 tx	10,000 tx
zkKYC/zkCBPR (Our Framework)	120.3 s	126.9 s	245.3 s	366.4 s
Quadrata [35]	2.0 s	21.1 s	222.1 s	2,311.7 s
Verite [16]	1.4 s	19.3 s	151.2 s	1,592.7 s
Zeeve ID / Privado [95]	48.7 s	471.3 s	4,912.3 s	51,670.0 s
Holonym [34]	2.3 s	23.4 s	310.7 s	3,308.2 s

5.4 Gas Consumption

We evaluated the *time* and *gas* consumption of smart contract functions. As mentioned in Section 4.4.3, our deployment and evaluation leveraged the local development environment Anvil [23] and a live Ethereum-based testnet deployment on Sepolia [20]. Gas consumption metrics offer valuable insights into the computational efficiency and resource usage of smart contract operations on the blockchain network. Table 5 presents a summary of gas consumption statistics gathered from our analysis of gas reports during testing on the Sepolia Ethereum testnet. These gas reports provide detailed data on gas consumption for each smart contract function, including the minimum, average, median, and maximum gas costs per function call. This information highlights the computational complexity and resource demands of individual operations within the smart contract.

5.5 End-to-End zkKYC/zkCBPR Workflow Evaluation

To complement the component-level benchmarks presented above, we conducted a lightweight end-to-end evaluation of the full zkKYC/zkCBPR workflow. The goal of this experiment is not to stress-test individual cryptographic primitives, but to empirically validate the feasibility and latency of the complete protocol pipeline described in Figures 6 and 7 under realistic execution assumptions.

We implemented an emulated deployment in which all stakeholders (CA, IE, sender bank, intermediary bank, and receiver bank) were instantiated as logically separate services on a single machine. Synthetic users were generated with unique DIDs and transaction histories, while smart contracts were executed against a local Ethereum development network (Anvil) to ensure consistency with on-chain execution semantics.

Each end-to-end run consisted of the following phases: (1) DID issuance and registration by the CA, (2) CSR submission and verification by an IE, (3) zkKYC proof generation and verification, (4) zkCBPR fund sufficiency proof generation by the sender's bank, (5) proof and certificate verification by the intermediary bank, (6) on-chain payment execution, and (7) MTR update following transaction settlement. Table 6 reports the average latency for each phase, averaged over five independent runs with identical transaction parameters. The results show that the dominant cost arises from local ZKP generation, while on-chain verification and settlement incur comparatively low latency. Importantly, the full end-to-end workflow completes within a few minutes even under conservative assumptions, demonstrating the practical deployability of zkKYC/zkCBPR as a system-level protocol rather than a collection of isolated cryptographic components.

5.6 Transaction Fees and Contract Deployment Costs

Transaction fees and contract deployment costs are essential considerations for assessing the economic viability of our zkKYC/zkCBPR framework. Here are the costs of deploying smart contracts and executing transactions on the Sepolia Ethereum Testnet.

Table 5. **Gas Consumption with Estimated USD Cost.**

(The `addIntermediate`, `removeIntermediate`, `registerDID`, and `updateMerkleRoot` functions are in the `MerkleTreeWithDID` smart contract and the `initiatePayment` function is in the `CBPRPaymentSystem` smart contract.)

Function Name	min		avg		median		max		# calls
	Gas	USD (\$)	Gas	USD (\$)	Gas	USD (\$)	Gas	USD (\$)	
<code>addIntermediate</code>	23,680	1.42	30,650	1.84	31,500	1.89	48,580	2.91	10
<code>removeIntermediate</code>	37,980	2.28	43,617	2.62	43,711	2.62	52,799	3.17	10
<code>registerDID</code>	29,101	1.75	30,074	1.80	30,065	1.80	31,178	1.87	10
<code>updateMerkleRoot</code>	111,739	6.70	120,138	7.21	119,302	7.16	147,420	8.85	10
<code>initiatePayment</code>	33,827	2.03	38,827	2.33	37,927	2.28	42,301	2.54	12

- **MerkletreeWithDID Deployment Cost:** 0.00771135 ETH
- **CBPRPaymentSystem Deployment Cost:** 0.00262354 ETH
- **CBPRPaymentSystem Transaction Fee:** 0.000257135778070161 ETH

The fees for executing transactions and deploying smart contracts are minimal, making our zkKYC/zkCBPR framework economically viable. When quantified using a nominal equivalent of 1 ETH $\approx$ \$4,100 (August 2025), the total deployment cost for all necessary smart contracts is approximately \$42.37. This one-time cost becomes negligible when amortized across thousands of subsequent transactions. For instance, spread over 10^4 transactions, the amortized cost is only about \$0.004 per transaction.

As can be seen in Table 7, the transaction fee to transfer a value of \$1,000 in our zkKYC/zkCBPR framework was approximately \$1.05 on the Sepolia Ethereum testnet (August 2025). This fee, which is independent of the transaction value, stands in stark contrast to traditional financial systems. Bank wire transfers typically incur a fixed fee, while credit cards and payment gateways generally charge 1.5%–3.5% of the transaction value (often with an additional fixed fee), making the total cost for medium-to-large transactions significantly higher than \$1.05. Consequently, our zkKYC/zkCBPR framework holds a distinct cost advantage for medium-to-large value settlements. Therefore, zkCBPR provides a consistent order-of-magnitude cost reduction for medium-to-large value settlements relative to incumbent payment rails.

Moreover, the entire fee structure in our zkKYC/zkCBPR framework is fully transparent, allowing users to track and verify the costs at every stage of the process. This combination of low fees and transparency ensures predictability and eliminates hidden costs, offering a clear advantage over traditional payment systems.

5.7 DID and MTR On-chain Storage

Our framework adopts a minimalistic on-chain storage model, where only a DID and its corresponding MTR are stored. This design ensures cryptographic guarantees for data integrity and auditability, while keeping the storage footprint low. Experimental results show that storing 10^4 [DID, root] pairs occupies approximately 1 MB of on-chain space, and each update operation consumes around 41,809 gas. To accommodate larger user volumes, the system can horizontally scale by deploying multiple instances of the MerkleTreeWithDID contract, each responsible for a subset of identities. This strategy effectively distributes the storage and gas load across multiple contracts, mitigating the limitations of monolithic storage architectures.

6 Privacy-Preservation and Trust Relationships

Our zkKYC/zkCBPR framework demonstrates that identity authentication and financial transactions can be performed in a decentralized environment without requiring users to disclose raw personal information to any transaction participants. The only point at which a user's real-world identity is revealed is during the initial

Table 6. End-to-End zkKYC/zkCBPR Latency Breakdown in a Local Instance

Phase	Avg. Time (s)	Source / Notes
(1) DID issuance and registration	< 0.1	Local Anvil net (instant block)
(2) CSR verification and signing	< 0.1	Lightweight EdDSA signature (instant computation)
(3) zkKYC proof generation	62.0	See 10 tx baseline in Figure 8
(4) Fund sufficiency ZKP generation	58.3	Total 120.3s from Table 4 - zkKYC proof generation time of 62s
(5) Proof and certificate verification	0.124	Constant time verification (see Section 5.3)
(6) On-chain payment execution	< 0.1	Local Anvil net (instant block)
(7) MTR update	0.114	See hashing time for 10 tx in Figure 8
Total Time	120.5	

Table 7. Transaction Fee Comparison Across Common Payment Systems

Payment System	Fee Model	\$1000 Transaction Fee
Swift (via U.S. banks)	Often \$25–\$50 (bank dependent)	\$25–\$50
Bank of America (wire)	\$30 domestic; \$45 international	\$30–\$45
Visa (merchant processing)	1.5%–3.5% + fixed fee	\$15–\$35
PayPal (U.S. merchant)	2.99% + \$0.49	\$30.39
zkKYC/zkCBPR (our framework)	Value-independent; network dependent	≈\$1.05

enrollment with the CA, where a DID is generated and anchored on-chain. From that point forward, all subsequent interactions, such as credential verification, transaction authorization, and compliance checks, are all mediated through ZKPs and signed certificates rather than direct data sharing. This approach minimizes data exposure, enforces a least-privilege principle across all actors, and builds a trust model where sensitive information is cryptographically protected rather than institutionally guarded.

Here are the various relationships between each party and how those impact privacy and trust.

- (1) **User** ↔ **CA**: As the root trust authority of the system, the CA verifies the user’s true identity, which requires ensuring that the user identity is strictly verified and the corresponding DID is generated and then registered on the blockchain, making it publicly verifiable in the decentralized network but not involved in the subsequent transaction process.
- (2) **User** ↔ **IE**: When a user wishes to conduct a transaction, the user submits a CSR to the IE containing the user’s DID and transaction details. After verifying the CSR, the IE converts it into an official transaction certificate. In this process, the IE acts as a verifier to ensure that the transaction is carried out with the authorization of the legitimate user without obtaining the user’s personal information.
- (3) **IE** ↔ **CA**: The IE only handles information related to the transaction and does not have the user’s personal information. The CA is responsible for verifying the user’s identity and generating a DID but will not involve specific transaction data or interfere with the subsequent transaction process.
- (4) **User** ↔ **The intermediary bank**: In the interaction between the *sender’s bank* and the *intermediary bank*, the *sender* and the *sender’s bank* need to submit three key documents, including:
 - **MTR ZKP**: The first ZKP document submitted by the *sender’s bank*, proving that the IE has a complete transaction record and has verified the transaction.
 - **SufficientFunds ZKP**: The second ZKP document submitted by the *sender’s bank*, verifying the sufficiency of the transaction amount.
 - **Transaction Certificate**: The *sender’s bank* provides a transaction certificate generated by the IE as proof of the sender’s legitimacy of the transaction request.

Therefore, given the above trust relationships: (i) The IE does not know the user’s personal information and only processes data related to the transaction; (ii) None of the intermediary banks hold the user’s identity information and can only verify the ZKP file and transaction certificate provided by the IE; (iii) As the only entity holding the user’s personal information, the CA is responsible for the initial identity authentication and DID generation but does not participate in the specific transaction process or share the user’s transaction information with the IE.

6.1 Trust Assumptions

- **Honesty of CA**: Our system assumes that the CA behaves honestly when verifying user attributes and issuing initial DID credentials. A malicious or compromised CA could theoretically issue fraudulent identities, undermining the integrity of subsequent transactions. To mitigate this, we recommend a federated or consortium-based CA structure, wherein credential issuance requires multi-signature approval from

multiple independent authorities, minimizing the risk posed by a single malicious entity. Additionally, compromise of a user's private key could enable impersonation attacks. While key compromise handling is out of scope for this work, established DID mechanisms provide mitigation through key rotation, whereby a user can update or revoke compromised keys in the DID document without changing the DID itself. This preserves identity continuity and auditability while restoring control to the legitimate user.

- **Integrity of ZKP Circuits:** The security of our system depends on the integrity and correctness of ZKP circuits used to generate credential proofs. We assume the cryptographic primitives (zkSNARK, hash functions) are secure. The main threats are malicious users attempting to falsify compliance (mitigated by requiring CA-issued credentials and ZKPs) and malicious CAs or IEs (mitigated by oversight and the fact that verifiers check all proofs). Incorrectly designed circuits or compromised trusted setups could leak sensitive user information. Regular cryptographic audits and open-source verification of the ZKP circuits are necessary precautions to maintain circuit integrity.
- **Blockchain Integrity and Decentralization:** We assume that the underlying blockchain infrastructure hosting credential registries remains sufficiently decentralized and resistant to 51% attacks. If the blockchain security assumptions fail (e.g., due to centralized control or compromised nodes), stored credentials could become vulnerable to manipulation. Selecting well-established blockchain networks with proven decentralization (e.g., Ethereum mainnet or major Layer-2 solutions) mitigates this risk.

7 Customizability and Scalability

Our zkKYC/zkCBPR framework is highly customizable, allowing financial institutions to tailor elements such as privacy settings, smart contract logic, and compliance rules to meet specific regulatory requirements and market needs. For example, an institution could adjust the ZKP circuit to enforce region-specific KYC criteria or set different Merkle tree update frequencies based on their risk profile. This adaptability makes the framework suitable for small-scale remittance services and large-scale institutional transactions.

7.1 Customization Options

The following dimensions in our zkKYC/zkCBPR framework can be customized to suit specific use cases:

- (1) *Variable Proof Systems:* The framework can be configured to support multiple ZKP backends, including zkSNARKs [11], zkSTARKs [9], and Bulletproofs [12], allowing users to choose the protocol that best suits their needs, which ensures that organizations can implement the framework across a variety of blockchain ecosystems. This modularity further allows implementers to balance trade-offs between proof size, verification speed, transparency (trusted setup vs. trustless), and post-quantum security, ensuring compatibility with a wide range of blockchains.
- (2) *Accommodating Multiple Blockchain Networks:* Cross-chain interoperability is essential for CBPR scenarios, where sender and receiver banks may operate on different blockchains. Our zkKYC/zkCBPR framework can be extended to support other blockchains by incorporating appropriate cross-chain communication protocols or interoperability layers (e.g., IBC [37], Chainlink CCIP [13]), without requiring major changes to the core architecture, thereby enabling seamless credential verification and transaction settlement across heterogeneous networks.
- (3) *Configurable Privacy Controls:* The level of detail in a CSR can be adjusted to show only essential information, ensuring privacy while supporting secure and compliant transactions. For instance, deployments can choose to reveal only minimal compliance proofs (e.g., "KYC verified") while keeping other attributes hidden.

7.2 Scalability Considerations

The scalability of our zkKYC/zkCBPR framework primarily revolves around two core areas: ZKP computation efficiency and blockchain transaction throughput. Efficiency in ZKP generation and verification is crucial for supporting high transaction throughput. While recent advances in ZKP systems have reduced computational overhead [10, 41], generating cryptographic proofs for millions of everyday transactions remains non-trivial. Empirical benchmarks show that ZKP generation becomes increasingly time-consuming as the depth of the Merkle tree and transaction history grow [42, 62]. To maintain sub-second performance at scale, implementations must leverage hardware acceleration (e.g., GPUs/FPGAs), optimized ZKP circuits, and batching strategies.

The underlying blockchain infrastructure further limits scalability. For example, the Ethereum mainnet supports approximately 15-30 transactions per second [25], which is insufficient for high-volume CBPR transfers. To address this issue, our zkKYC/zkCBPR framework minimizes on-chain data by storing only the MTR and aggregated commitments, while retaining a detailed transaction log off-chain with verifiable links. In production-scale deployments, layer-2 solutions such as rollups [14, 43] and validium [26], or even dedicated permissioned chains, can significantly increase throughput without compromising security and privacy.

Furthermore, our framework's modular design can encompass multiple blockchain networks, each dedicated to DID management, sender banking operations, and receiver settlement. While this approach improves separation of concerns, it also introduces challenges in inter-chain coordination. Cross-chain communication protocols must be optimized to ensure low-latency messaging and rapid settlement finality, in such scenarios.

Finally, as central banks worldwide accelerate their exploration and pilot deployments of CBDCs [5, 24], it is essential that our zkKYC/zkCBPR framework remain adaptable to these emerging paradigms. To that end, further stress testing, interoperability assessments, and conformance checks against upcoming specifications are necessary to guarantee that the framework can seamlessly integrate with next-generation payment infrastructures while preserving its privacy and compliance guarantees.

7.3 Deployment Considerations

While our prototype demonstrates the technical feasibility of decentralized, privacy-preserving compliance, its practical adoption will depend on integration with existing institutional and regulatory ecosystems. In a real-world deployment, CAs could be operated by a consortium of regulated entities such as licensed banks, government agencies, or trusted compliance providers, following a governance framework akin to the Sovrin model [59, 93]. IEs responsible for certificate issuance and MTR updates could be run by financial institutions, decentralized service providers, or other regulated entities participating in the ecosystem. All issued DIDs and credential metadata would be anchored on a public blockchain, such as Ethereum, using a smart contract-based registry, which maps each DID to a corresponding MTR and public key. This enables any verifier or smart contract to validate a user's proof of compliance without relying on centralized data access. These components collectively provide a pathway from prototype to production-ready infrastructure while preserving interoperability, auditability, and decentralized trust.

7.3.1 Challenges and Potential Solutions. While our zkKYC/zkCBPR framework provides strong privacy guarantees and operational efficiency, its real-world deployment in CBPR systems will likely face several challenges.

One important barrier stems from market resistance and institutional inertia. Many traditional financial institutions are risk-averse and accustomed to centralized architectures where all transaction data is auditable. Migrating to a cryptographic, decentralized compliance paradigm introduces operational and cultural friction. Hybrid deployment models, where ZKPs complement rather than replace traditional workflows, could act as a transitional solution. Additionally, public-private partnerships and well-documented pilot programs will be essential to demonstrate feasibility and build institutional trust.

Regulatory barriers are also a major challenge. Financial regulators operate under strict oversight to ensure AML, CTF, and consumer protection. While the opacity of ZKPs can protect privacy, it may conflict with regulators' expectations for auditability and traceability. In our zkKYC/zkCBPR framework, the CA retains the binding between a user's DID and their real-world identity, while participating banks and IEs retain the underlying transaction records. This separation preserves privacy on-chain while still enabling auditability and regulatory investigations when legally required.

While this separation of responsibilities supports auditability, regulatory compliance requirements vary significantly across jurisdictions. As a result, systems intended for global deployment must accommodate differing expectations around disclosure, traceability, and reporting. For example, the Financial Action Task Force (FATF) Travel Rule requires virtual asset service providers to transmit identifying information about both senders and receivers alongside transactions [32], while the European Union's Fifth and Sixth Anti-Money Laundering Directives (AMLD5/6) expand AML obligations to cryptocurrency exchanges, mandate beneficial ownership registries, and broaden the scope of predicate offenses for money laundering [29, 30]. These evolving requirements highlight the tension between privacy-preserving technologies and demands for traceability.

Addressing these barriers requires active regulatory engagement, transparent audit mechanisms that coexist with privacy protections, and the possible development of regulator-specific verification portals to balance regulation and confidentiality. Therefore, our zkKYC/zkCBPR framework cannot completely replace the existing global CBPR framework, but it can serve as a complementary option to strengthen user privacy protection in CBPR.

Interoperability and standardization pose additional challenges. Although DID standards such as W3C DIDs [17] and VCs [18] have gained traction, integration with existing financial infrastructures and across different blockchains remains limited. Without alignment on technical and procedural standards, the ecosystem risks fragmentation. This underscores the need for interoperable middleware, cross-chain communication protocols, and active participation in global standardization efforts.

Finally, user trust and adoption remain pivotal. For both individuals and institutions, ZKPs may appear opaque and difficult to understand, potentially impeding usage. To ensure usability, the system must incorporate transparent governance, intuitive user interfaces, and legal clarity. Institutional endorsement, particularly from central banks or trusted regulators, could play a major role in bolstering user confidence.

8 Conclusion

This paper introduces zkKYC/zkCBPR, a pair of complementary frameworks designed to set a new standard for DID and privacy-preserving regulatory compliance. By seamlessly integrating SSI, DID, and ZKPs, our zkKYC/zkCBPR framework enables users to cryptographically prove their compliance (e.g., KYC verification or financial solvency) without exposing sensitive personal data. In doing so, we resolve the long-standing tension between privacy and institutional trust, offering a scalable, verifiable, and trust-minimized alternative to legacy centralized systems that are vulnerable to data breaches, inefficiencies, and excessive surveillance.

Interoperability with emerging DID and ZKP systems is a core design consideration of zkKYC/zkCBPR. Because the framework relies on W3C-conformant DIDs, VCs, and standard zkSNARK verification interfaces, it can be readily adapted to newer identity ecosystems, including ZK-native rollups, modular identity layers, and evolving credential formats. For example, alternative proof systems or DID methods can be integrated without modifying the on-chain verification logic, provided they expose compatible verification keys and public inputs. This flexibility positions zkKYC/zkCBPR as a future-proof trust infrastructure capable of evolving alongside the decentralized web.

Our approach fundamentally rebalances control in the decentralized web, transferring authority from opaque intermediaries to user-controlled VCs and smart contracts. It ensures strong cross-chain interoperability, immutable

auditability, and cost-effective deployment, making it indispensable for high-stakes applications in finance, e-commerce, and Web3 ecosystems. Through rigorous evaluation, including ZKP performance benchmarks, gas cost profiling, and scalability assessments, we demonstrate that the framework achieves sub-two-minute proof generation even for large datasets, with transaction costs as low as \$1.05. These results underscore that zkKYC/zkCBPR is not merely a conceptual proposal but a practical, high-performance solution ready for real-world deployment, with future directions including integration with CBDCs, alignment with global regulatory frameworks, and adoption in domains such as cross-border remittances, DeFi onboarding, and healthcare identity verification. Thus, our evaluation demonstrates not only cryptographic feasibility but also system-level deployability.

Overall, our zkKYC/zkCBPR framework paves the way for decentralized compliance infrastructures that reconcile privacy with regulation, a critical challenge for the future of Web3 and digital finance. By bridging technical innovation with regulatory realities, our zkKYC/zkCBPR framework positions itself as a foundational layer for privacy-preserving trust in next-generation financial systems. Although our system is implemented and evaluated in the context of KYC requirements for CBPR, including AML and CTF, we argue that this design could be generalized to a wide range of decentralized web applications beyond finance.

Resources:

- Our source code is available at https://github.com/brains-group/zkKYC_and_zkCBPR under Apache 2.0 license.
- A video demonstration of the framework is available at <https://youtu.be/YxnflidtYOM>.

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